1

a

Entity Sets

- Account
- Branch
- Customer
- Loan

Relations

- Account branch

>> Many Accounts to One Branch with all Accounts needing to have a branch

- Loan branch

>> Many Loans to One Branch with all Loans needing to have a branch

- Depositor

>> Accounts to Customer

- Borrower

>> Customer to Loan

b

Entities

Account(Account_No, Balance, B_name)

Branch(B_name, B_city, Assets)

Customer(C_name, C_city, C_street)

Loan(Loan_No, Amount, B_name)

Relations

Depositor(Account_No,C_name)

Borrower(C_name,Loan_No)

c

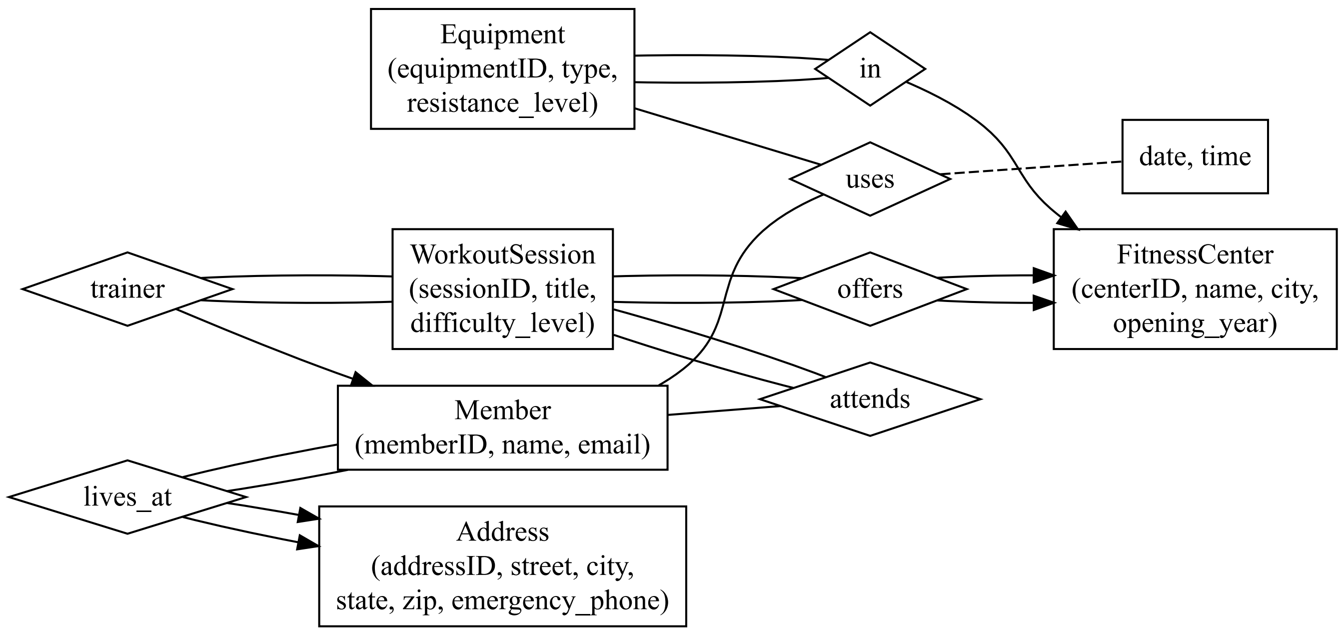
In Account and Loan, B_name should have a foreign key constraint and a not null constraint

In Depositor, Account_No and C_name should have foreign key constraints

In Borrower, Loan_No and C_name should have foreign key constraints

2

a



Assumptions

1. All Equipments must be in a Fitness Center
2. All Fitness Centers must offer at least one Workout Session
3. All Workout Sessions must have one trainer, at least one member attending, and be in one fitness center
4. All Members have one Address
5. All Addresses correspond to at least one Member
6. Member can use a piece of Equipment more than once so a date and time was created

b

Equipment(<u>equipmentID</u>, type, resistance_level, centerID)

> FK centerID contains PKs from FitnessCenter

FitnessCenter(<u>centerID</u>, name, city, opening_year, sessionIDs)

> sessionIDs contains a non empty list of WorkoutSession PKs (FK Constraint) and ALL sessions are covered

WorkoutSession(<u>sessionID</u>, title, difficulty_level, trainerID, centerID)

> FK centerID contains PKs from FitnessCenter and ALL centers are covered

> FK trainerID contains PKs from Member

Attends(<u>sessionID, memberID</u>)

> FK sessionID contains PKs from WorkoutSession

> FK memberID contains PKs from Member

Member(memberID, name, email, addressID)

> FK addressID contains PKs from Address and ALL Address entries are covered

Address(addressID, street, city, state, zip, emergency_phone)

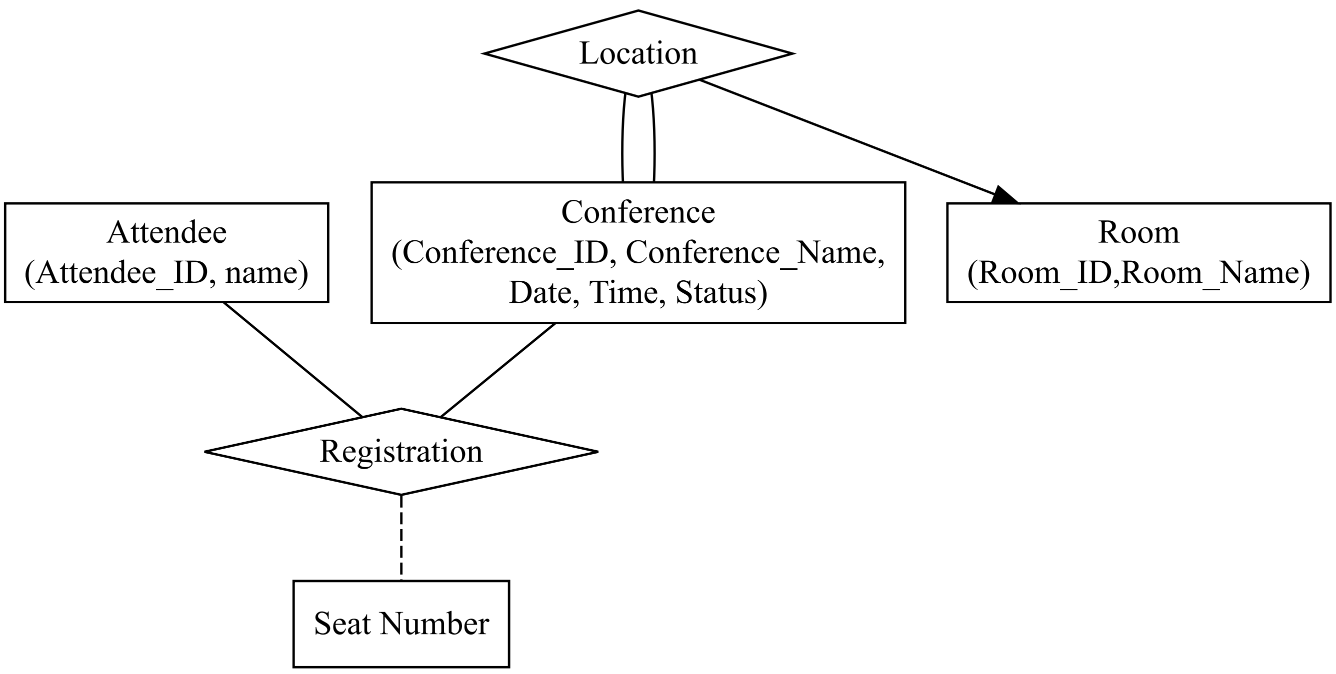
Uses(equipmentID, memberID, date,time)

> FK equipmentID contains PK from Equipment

> FK memberID contains PKs from Member

3

a



Assumptions

1. Seats are numbered specifically for an event
2. Attendee only has one seat for an event
3. Conference only has one room
4. Conference_ID References a Unique instance of a conference (ie if a conference has multiple schedules they have different ids despite having the same name but different times)
5. Attendee doesn't need to be regestired to a conference (could be in system from a previous year, maybe they didn't complete all registration steps etc)
6. Not all conferences need to have an attendee

b

Attendee(Attendee_ID, name)

Conference(Conference_ID, Conference_Name, Ddate, Time, Status, Room_ID, Room_Name)

Registration(Attendee_ID,Conference_ID, Seat Number)

Constraints

Registration should have foreign key constraints on Attendee_ID and Conference_ID and the two should have key constraints as well

Attendee and Conference should have key constraints on Attendee_ID and Conference_ID respectively